Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam D · Form 1: Multi-step reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. A horizontal launch passes through a window a distance D from the table and h below launch. Which launch speed hits the window center?

(A) Dsqrt(g/(2h))

(B) Dsqrt(2h/g)

(C) sqrt(2gh)

(D) gD/h

2. Two balls are launched simultaneously from the same height with the same positive initial vertical velocity. B has 11 times A’s horizontal velocity. Both land on the same ground without drag. Compare flight times and ranges.

(A) Both B’s flight time and range are 11 times A’s.

(B) Flight times are equal; B’s range is 11 times A’s.

(C) B takes 11 times as long; ranges are equal.

(D) Both flight times and ranges are equal.

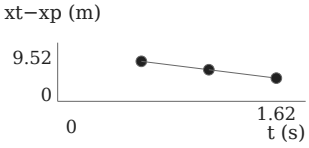
3. A projectile moves horizontally at +6 m/s while a target travels at +2 m/s. Track their horizontal separation during flight. t (s) -> xt-xp (m): (0.62, 9.52); (1.12, 7.52); (1.62, 5.52). A logger displays twice each true horizontal reading. What slope appears on the displayed axes?

(A) -2.8

(B) -1.2

(C) -0.4

(D) -2



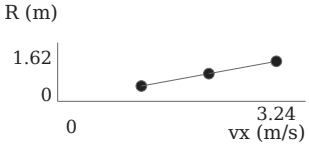
4. A ball launches horizontally from a 1.25 m table. Use g=10 m/s² and neglect drag. Quadruple the table height. Under the revised model, what is Y at X=2.24?

(A) 2.24

(B) 2.345

(C) 2.03

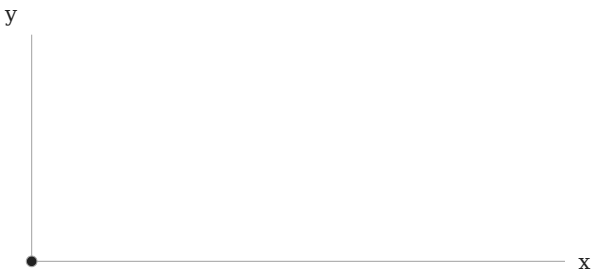
(D) 2.555



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A ball is launched from ground at 21 m/s, 35° above horizontal, and lands on the same ground. Ignore drag.

a. Draw the initial velocity vector and its horizontal and vertical components. **(1 pt)**



b. Find flight time, range, and velocity and acceleration at the top. **(3 pt)**

6. A map route starts at east coordinate 2 m and runs at 60° north of east. d (m) -> east (m): (1.24, 2.62); (2.24, 3.12); (3.24, 3.62).

a. Build the model for eastward projection from physics and predict both slope and intercept. **(2 pt)**

b. Rotate the route to 120° north of east at the same starting point. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A ball is launched from height h at speed u and angle θ above horizontal. Derive positive time to ground impact and horizontal range. **(3 pt)**